Darshan UNIVERSITY

योग: कर्मसु कौशलम्

Python Programming - 2301CS404

Lab - 6

Roll No. : 495

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01) WAP to find sum of all the elements in a List.

In [1]:
```
l = list(map(int, input("Enter numbers separated by space: ").split()))
total_sum = sum(l)
print("Sum of all elements:", total_sum)
```

```
Enter numbers separated by space:  10
[10]
Sum of all elements: 10
```

02) WAP to find largest element in a List.

In [ ]:
```
l2 = [1,2,4,5,100,45]

largest = max(l2)

print("Largest of all elements:",largest)
```

03) WAP to find the length of a List.

In [9]:
```
l3 = [1,2,3,4,5]

length = len(l3)

print("Length of the list is:",length)
```

```
Length of the list is: 5
```

04) WAP to interchange first and last elements in a list.

In [2]:
```
lst = list(map(int, input("Enter elements separated by space: ").split()))

lst[0], lst[-1] = lst[-1], lst[0]

print("Updated List:", lst)
```

```
Enter elements separated by space:  1 2 3 4 5 6 7 8
Updated List: [8, 2, 3, 4, 5, 6, 7, 1]
List after swapping first and last elements: ['4', '2', '3', '1']
```

05) WAP to split the List into two parts and append the first part to the end.

In [14]:
```
lst = list(map(int, input("Enter elements: ").split()))

n = int(input("Enter position to split: "))

result = lst[n:] + lst[:n]


print("Updated List:", result)
```

```
Enter elements:  1 2 3 4 5 6 7 8
Enter position to split:  3
Updated List: [4, 5, 6, 7, 8, 1, 2, 3]
List after splitting and rearranging: ['5', '6', '7', '1', '2', '3', '4']
```

06) WAP to interchange the elements on two positions entered by a user.

In [13]:
```
list = input("Enter elements separated by space: ").split()

pos1 = int(input("Enter first position to swap: "))
pos2 = int(input("Enter second position to swap: "))

list[pos1], list[pos2] = list[pos2], list[pos1]

print("List after swapping:",list)
```

```
Enter elements separated by space:  1 2 3 4 5
Enter first position to swap:  1
Enter second position to swap:  2
List after swapping: ['1', '3', '2', '4', '5']
```

07) WAP to reverse the list entered by user.

In [4]:
```
li = input("Enter elements separated by space: ").split()
rev_li = li[::-1]
print("Reversed List:", rev_li)
```

```
Enter elements separated by space:  2 3 4 5 6
Reversed List: ['6', '5', '4', '3', '2']
Reversed List: ['4', '3', '2']
```

08) WAP to print even numbers in a list.

In [4]:
```
lst = list(map(int,input("Enter elements: ").split()))

for i in lst:
    if i % 2 == 0:
        print(i)
```

```
Enter elements:  1 2 3 4 5 6 7 8
2
4
6
8
```

09) WAP to count unique items in a list.

In [8]:
```
li = [1, 2, 2, 3, 4, 4, 5, 6, 6, 7]

unique_items = set(li)

count = len(unique_items)

print("Number of unique items:", count)
print("Unique items are:", unique_items)
```

```
Number of unique items: 7
Unique items are: {1, 2, 3, 4, 5, 6, 7}
```

10) WAP to copy a list.

In [22]:
```
og_list = [10, 20, 30, 40, 50]

copied_list = og_list.copy()

#print("Original List:", og_list)
print("Copied List:", copied_list)
```

```
Copied List: [10, 20, 30, 40, 50]
```

11) WAP to print all odd numbers in a given range.

In [6]:
```
start = int(input("Enter the start of range: "))
end = int(input("Enter the end of range: "))

for i in range(start, end + 1):
    if i % 2 != 0:
        print(i)
```

```
Enter the start of range:  1
Enter the end of range:  10
1
3
5
7
9
1
3
5
7
9
```

12) WAP to count occurrences of an element in a list.

In [29]:
```
numbers = [2, 3, 4, 2, 5, 2, 7, 2]
a = []

for i in numbers:
    if i not in a:
        count = numbers.count(i)
        a.append(i)
        print("The element",i, "appears", count, "times.")
```

```
The element 2 appears 4 times.
The element 3 appears 1 times.
The element 4 appears 1 times.
The element 5 appears 1 times.
The element 7 appears 1 times.
```

13) WAP to find second largest number in a list.

In [30]:
```
numbers = [10, 20, 4, 45, 99, 67]

numbers.remove(max(numbers))

second_largest = max(numbers)

print("Second largest number is:", second_largest)
```

```
Second largest number is: 67
```

14) WAP to extract elements with frequency greater than K.

In [8]:
```
numbers = list(map(int, input("Enter numbers separated by space: ").split()))
print(numbers)
k = int(input("Enter the value of K: "))

freq = {}

for num in numbers:
    if num in freq:
        freq[num] += 1
    else:
        freq[num] = 1

result = []

for key in freq:
    print(key)
    if freq[key] > k:
        result.append(key)

print("Elements with frequency greater than", k, "are:", result)
```

```
Enter numbers separated by space:  1 2 2 3 3 3 4 4 4 4
[1, 2, 2, 3, 3, 3, 4, 4, 4, 4]
Enter the value of K:  2
```

```
1
2
3
4
Elements with frequency greater than 2 are: [3, 4]
```

15) WAP to create a list of squared numbers from 0 to 9 with and without using List Comprehension.

In [32]:
```
#squares = []
#for i in range(10):
    #squares.append(i * i)
#print("Squares without list comprehension:", squares)

og_list = [1,2,3,4,5,6,7,8,9]
l = [i**2 for i in og_list]
print(l)
```

```
[1, 4, 9, 16, 25, 36, 49, 64, 81]
```

16) WAP to create a new list (fruit whose name starts with 'b') from the list of fruits given by user.

In [10]:
```
fruits = input("Enter fruit names separated by space: ").split()

b_fruits = []

for fruit in fruits:
    if fruit.lower().startswith('b'):
        b_fruits.append(fruit)

print("Fruits starting with 'b':", b_fruits)
```

```
Enter fruit names separated by space:  banana apple guava
Fruits starting with 'b': ['banana']
Fruits starting with 'b': ['banana']
```

17) WAP to create a list of common elements from given two lists.

In [11]:
```
list1 = list(map(int, input("Enter numbers separated by space1: ").split()))
list2 = list(map(int, input("Enter numbers separated by space2: ").split()))

common = []

for i in list1:
    if i in list2:
        common.append(i)

print("Common elements:", common)
```

```
Enter numbers separated by space1:  1 2 3 4 5 6 7
Enter numbers separated by space2:  1 2 9 8 7 10 11
Common elements: [1, 2, 7]
```

In [ ]: